

Judul

David Folio[✉], *Member, IEEE* and John Doe

Abstract—This document describes the most common article elements and how to use the `quarto-ieee` class with Pandoc/Quarto-Markdown to produce files that are suitable for submission to IEEE journals.

`quarto-ieee` can produce conference, journal, and technical note (correspondence) papers with a suitable choice of class options. It intends to generate PDF and HTML outputs that closely mimick what IEEE would generate.

Index Terms—IEEE, IEEEtran, journal, Quarto, Pandoc, template

I. INTRODUCTION

THIS file is intended to serve as a “sample article file” for IEEE journal papers produced with (Pandoc/Quarto)-Markdown using `IEEEtran.cls` version 1.8b and later for the PDF output. It is based on `bare_jrnl_new_sample4.tex` provided by IEEE Publication Technology, Staff and available from <https://template-selector.ieee.org/>. The most common elements are covered in the simplified and updated instructions in `New_IEEEtran_how-to.pdf`. For less common elements you can refer back to the original `IEEEtran_HOWTO.pdf`. It is assumed that the reader has a basic working knowledge of $\text{\LaTeX}$ [1] and of (Pandoc/Quarto)-Markdown [2], [3] markup.

1) *Figures*: An image with nonempty alt text will be rendered as a figure with a caption with Pandoc and Quarto. Quarto includes a different features to simplify the use of figures and subfigures. Here, it is recommended to use `div` block with `#fig-` label to embed your Figures.



Fig. 1: An example of figure.

The figures is cross-referenced as Fig. 2 and even the sub-figures as Fig. 2b.

2) *Tables*: Similarly, many kind of tables may be used with Pandoc and Quarto. The latter also includes different features to simplify the table output. To make tables cross-referenceable use a label with a `#tbl-` prefix. However, it is recommended to avoid using the commonly

The `quarto-ieee` template is freely available under the MIT license on github: <https://github.com/dfolio/quarto-ieee>.

David Folio is with Laboratoire Prisme, INSA Centre Val de Loire, Bourges, 18800 France Corresponding author: david.folio@insa-cvl.fr

Unknown affiliation

John Doe is with Anonymous University

Template created June 23, 2023; revised r
format(Sys.Date(), format='%B %d, %Y').

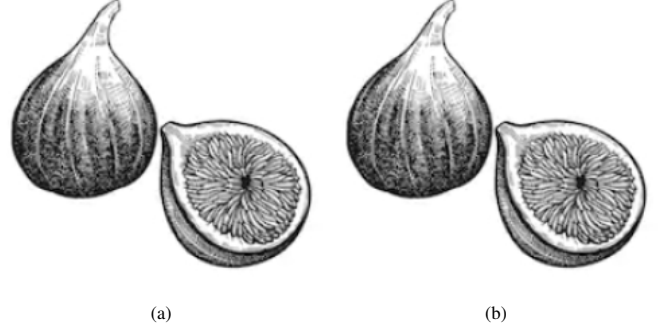


Fig. 2: An example with sub-figure.

used single Markdown table known as a ‘pipe table’. In fact, Pandoc Markdown uses the $\text{\LaTeX}$ `longtable` package, which does not support the two-column mode, which is required for most IEEEtran journals. `quarto-ieee` uses a hack to temporarily switch to one-column mode. However, this hack may break the page layout. To overcome this issue, a basic way is to use code cells (as for Table II). Quarto is a multi-language and it uses Knitr to execute R code and can execute Python code blocks within Markdown.

Table I: Main Caption

(a) First Table			(b) Second Table		
Col1	Col2	Col3	Col1	Col2	Col3
A	B	C	A	B	C
E	F	G	E	F	G
A	G	G	A	G	G

The Tables are cross-referenced as Table I for details, especially Table Ib. There is also Table II.

Table II: A table

Col1	Col2	Col3
A	D	G
B	E	H
C	F	I

A. Bibliography

IEEE journal should normally use IEEEtran¹ $\text{\LaTeX}$ style. Nevertheless, Pandoc and Quarto do support $\text{\LaTeX}$ with `natbib` or `biblatex`. However, neither is officially recommended for normal IEEE use. For this reason, `quarto-ieee` uses `citeproc` with the `ieee` CSL style sheet.

II. CONCLUSIONS

The conclusion goes here.

¹IEEEtran BibTeX style support page is: <http://www.michaelshell.org/tex/ieee/IEEEtran/bibtex/>

ACKNOWLEDGMENT

This should be a simple paragraph before the References to thank those individuals and institutions who have supported your work on this article.

APPENDIX AN APPENDIX

Use `[]{\appendix options="An Appendix"}` markup if you have a single appendix. `IEEEtran` state that to do not use `\section{}` anymore after `\appendix`.

REFERENCES

- [1] F. Mittelbach and U. Fischer, *The LaTeX companion*, 3rd ed. Addison Wesley Professional, 2023.
- [2] J. MacFarlane, A. Krewinkel, and J. Rosenthal, "Pandoc." [Online]. Available: <https://github.com/jgm/pandoc>
- [3] J. J. Allaire, C. Teague, C. Scheidegger, Y. Xie, and C. Dervieux, "Quarto." Jan-2022 [Online]. Available: <https://github.com/quarto-dev/quarto-cli>



David Folio Use `IEEEbiography` with `figure` as option and the author name as the argument followed by the biography text.

John Doe Use `IEEEbiographynophoto` and the author name as the argument followed by the biography text.